



E-COMMERCE ADMIN LOGIN WITH DECOY PROTECTION



A PROJECT REPORT

Submitted by

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*in partial fulfillment of the requirements for the award degree of
Bachelor of Engineering*

CSB1332 - DESIGN PROJECT
**DEPARTMENT OF COMPUTER SCIENCE AND
ENGINEERING**

K. RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(AUTONOMOUS)
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BONAFIDE CERTIFICATE

The work embodied in the present project report entitled
“E - COMMERCE ADMIN LOGIN WITH DECOY PROTECTION ”
has been carried out by the students **ROHINI D, SHEEBA S, SINDHUJA S,
SRINITHI D.** The work reported herein is original and we declare that the
project is their own work, except where specifically acknowledged, and has not
been copied from other sources or been previously submitted for assessment.

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ABSTRACT

The rapid growth of e-commerce platforms has significantly increased the demand for strong E-commerce Security and Admin Login Protection mechanisms to safeguard sensitive business and customer data. Traditional authentication systems that rely only on usernames and passwords are highly vulnerable to cyber threats such as brute-force attacks, phishing, and unauthorized access. To address these issues, the proposed system E-Commerce Admin Login with Decoy Protection introduces an advanced Secure Authentication mechanism using OTP Authentication along with Honeywords and Honeytrap techniques. If the correct password is entered, the admin gains access to the real dashboard. If a honeyword is used, the intruder is redirected to a Decoy Dashboard, which behaves like a real system but secretly monitors and records attacker activities. This method enables effective Cybersecurity Intrusion Detection while protecting the actual system and database from potential threats. The proposed system is implemented using HTML, CSS, and JavaScript, along with E-Mail JS for OTP generation and alert notifications. It includes modules such as Login Portal, Authentication System, Password Recovery, Real Dashboard, Decoy Dashboard, and Alert System. Overall, this approach provides layered Web Security, ensuring safe login, intrusion detection, and secure data management for modern e-commerce applications.

Keywords: E-Commerce Security, Admin Login Protection, Honeywords and Honeytrap OTP Authentication, Decoy Dashboard, Cybersecurity Intrusion Detection, Secure Authentication, Web Security

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SIGNATURE

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LIST OF ABBREVIATIONS

CSS	-	Cascading Style Sheets
HTML	-	Hyper Text Markup Language
OTP	-	One-Time Password
EMAILJS	-	Email JavaScript Service
SQL	-	Structured Query Language
DB	-	Database
API	-	Application Programming
UAT	-	User Acceptance Testing
OS	-	Operating System

CHAPTER 1

INTRODUCTION

1.1 DESCRIPTION

E-Commerce platforms are widely used for buying and selling products online, and they require secure admin panels to manage products, users, payments, and orders. The admin panel is the most sensitive part of an e-commerce system because it controls the entire business data and operations. Traditional admin login systems mostly depend on username and password authentication, which makes them vulnerable to cyber-attacks such as brute-force attacks, phishing, and unauthorized access.

To improve the security of admin login systems, modern cybersecurity techniques such as OTP authentication, honeywords, and honeypots are used. Honeywords are fake passwords stored in the system to detect attackers, while honeypots are fake environments used to trap hackers and monitor their activities. By combining these techniques, it is possible to detect unauthorized access and protect the real system from cyber threats. Attackers can easily attempt multiple login combinations and gain access to confidential data.

The proposed system, Shop Nova Admin Panel with Decoy Protection, provides a secure login environment using OTP verification and decoy dashboard protection. If a real admin logs in, access is granted to the real dashboard, while attackers are redirected to a fake dashboard and their activities are recorded. This helps in protecting sensitive data and improving overall system security.

1.2 PROBLEM STATEMENT

The main objective of this project is to enhance the security of the E-commerce Admin Panel by implementing advanced authentication and protection mechanisms. The system aims to ensure that only authorized users can access the real admin dashboard while preventing unauthorized access attempts. One of the primary objectives is to implement multi-layer authentication, combining username-password verification with OTP (One-Time Password) validation to strengthen user identity verification. This reduces the risk of unauthorized access even if login credentials are compromised.

In the professional domain, employees and working individuals can use the application to monitor daily tasks, improve work consistency, and enhance overall productivity. It is also widely applicable in the health and wellness sector, where users can track habits such as exercise, diet, hydration, sleep patterns, and mental health activities. This helps in promoting a healthier and more balanced lifestyle.

The application also plays a significant role in self-development and behavioral improvement, as it encourages users to maintain discipline, set achievable goals, and monitor progress over time. By providing visual insights and progress tracking, the system helps users identify patterns and improve their daily routines effectively. From a technological perspective, the project falls under the cloud computing domain, as it utilizes Microsoft Azure for deployment, data storage, and scalability. This ensures secure data handling, high availability, and remote accessibility. The system also belongs to the Software as a Service (SaaS) category, where users can access the application through a web browser without needing complex installations.

Furthermore, the application can be extended to domains such as organizational management, fitness tracking systems, and goal management platforms, making it versatile and adaptable. Overall, the project demonstrates the practical implementation of web technologies and cloud services in solving real-life productivity challenges.

1.3 CYBER SECURITY

Cybersecurity is the field that focuses on protecting computer systems, networks, and data from unauthorized access, attacks, and damage. It ensures that sensitive information such as user details, passwords, and payment data remains safe and secure. In today's digital world, applications like e-commerce websites are common targets for hackers. Cybersecurity helps prevent threats such as data breaches, phishing attacks, malware, and unauthorized login attempts. It uses different techniques like encryption, authentication, firewalls, and monitoring systems to protect both users and organizations.

Cybersecurity is applied to secure the admin panel of an e-commerce system. It not only verifies user credentials but also uses advanced techniques like decoy login protection, honeypots, and honeywords to detect and trap attackers. This helps in preventing misuse of stolen credentials and keeps the real system and database protected.

1.3.1 Web Application Security Domain

The Web Application Security domain focuses on protecting websites and web-based applications from cyber-attacks and vulnerabilities. It ensures that important components such as login pages, admin panels, APIs, and databases are secure from threats like hacking, SQL injection, cross-site scripting (XSS), and unauthorized access.

This domain involves implementing secure coding practices, input validation, encryption, and proper session management to prevent attackers from exploiting weaknesses in the application. It also includes techniques such as authentication mechanisms, access control, and activity monitoring to ensure that only legitimate users can access sensitive features.

1.3.2 Authentication and Access Control Domain

The Authentication & Access Control domain focuses on verifying the identity of users and ensuring that only authorized individuals can access specific resources or perform certain actions within a system. Authentication is the process of confirming who the user is, typically using methods such as usernames and passwords, one-time passwords (OTP), or multi-factor authentication (MFA).

This domain implements different security mechanisms such as role-based access control (RBAC), session management, password policies, and account lockout features to prevent unauthorized access. It also helps in reducing risks like brute-force attacks, credential theft, and privilege misuse.

1.3.3 Cyber Deception (Honeypot Techniques) Domain

The Cyber Deception domain focuses on protecting systems by misleading attackers and diverting them away from real assets. Instead of only blocking attacks, this approach creates realistic fake environments, known as honeypots, to attract and trap malicious users. These decoy systems imitate real login pages, admin dashboards, or databases, making attackers believe they have gained access to a genuine system.

Once an attacker interacts with the decoy environment, their activities are carefully monitored and recorded. This helps in identifying attack methods, detecting suspicious behavior, and gathering useful information about potential threats. Unlike traditional security systems that only alert or block, cyber deception actively engages attackers while keeping the real system completely safe.

This domain includes techniques such as honeypots, honeywords (decoy passwords), fake servers, and shadow databases. These methods reduce the risk of real data exposure and provide an additional layer of security beyond standard authentication. In your project, Cyber Deception is implemented by creating a fake admin login and dashboard.

1.3.4 AI/ML Domain

The Artificial Intelligence (AI) and Machine Learning (ML) domain focuses on building systems that can learn from data, identify patterns, and make decisions with minimal human intervention. AI provides the overall intelligence, while ML is a subset that enables systems to improve their performance automatically by analyzing data. In cybersecurity, AI and ML are used to detect unusual behavior, identify threats, and respond to attacks in real time. These technologies can analyze login patterns, user behavior, and system activities to distinguish between normal users and potential attackers.

They help in detecting anomalies such as repeated failed login attempts, unusual access times, or suspicious actions. AI and ML concepts can be applied to enhance security by analyzing login behavior and identifying suspicious activities. The system can learn from past data to detect attackers more accurately and trigger the decoy login mechanism when abnormal patterns are found. This makes the security system more intelligent, adaptive, and effective against evolving cyber threats.

1.4 AIM AND OBJECTIVE

The aim of this project is to strengthen the security of an e-commerce admin system by developing an advanced decoy login protection mechanism. The system integrates honeywords (decoy passwords) and honeypot-based fake admin dashboards to detect and respond to unauthorized access attempts. Instead of only allowing or denying access, the system intelligently identifies suspicious login behavior and redirects attackers to a controlled decoy environment. This approach helps in preventing direct access to the real admin panel and ensures the protection of sensitive data such as customer information, order details, and payment records. The project also aims to provide proactive security by not only detecting threats but also misleading and monitoring attackers to improve overall system defense.

The objectives of this project are to design and implement a secure and user-friendly admin authentication system for an e-commerce platform. The system aims to incorporate honeyword-based password security to detect unauthorized login attempts by using decoy passwords. It also focuses on developing a realistic decoy admin dashboard using honeypot techniques to mislead attackers and prevent them from accessing the actual system. Additionally, the project aims to detect suspicious login behavior in real time and respond appropriately by redirecting unauthorized users to the fake environment. Another objective is to monitor and log attacker activities such as login attempts and actions performed for analysis.

This domain implements different security mechanisms such as role-based access control (RBAC), session management, password policies, and account lockout features to prevent unauthorized access. It also helps in reducing risks like brute-force attacks, credential theft, and privilege misuse.

1.5 SCOPE OF THE PROJECT

The scope of this project is to develop a secure e-commerce admin login system that protects against unauthorized access using decoy-based security techniques. It mainly focuses on strengthening the admin authentication process by integrating honeywords and honeypot mechanisms to detect and handle suspicious login attempts. The system is designed to safeguard sensitive information such as user data, order details, and payment records by preventing attackers from accessing the real admin panel.

This project covers the design and implementation of a login interface, password verification using honeywords, and a decoy admin dashboard that acts as a trap for attackers. It also includes monitoring and logging of attacker activities for analysis and improving system security.

The system is designed to safeguard sensitive information such as user data, order details, and payment records by preventing attackers from accessing the real admin panel. This project covers the design and implementation of a login interface, password verification using honeywords, and a decoy admin dashboard that acts as a trap for attackers. It also includes monitoring and logging of attacker activities for analysis and improving system security. The solution is intended to be practical and can be applied to real-world e-commerce platforms to enhance their security.

CHAPTER 2

LITERATURE SURVEY

2.1 H. B. PATEL, E-COMMERCE SECURITY THREATS AND DEFENSES [1]

The analysis focuses on the increasing number of cyber attacks targeting online business systems and highlights the need for strong authentication and protection mechanisms. It explains that e-commerce platforms manage large volumes of sensitive information such as customer data, payment details, and business transactions, making them highly attractive targets for cyber criminals.

Conventional security systems that depend only on username and password authentication are no longer sufficient to safeguard modern web applications from sophisticated threats. The evaluation describes common attack methods such as phishing, brute-force attacks, SQL injection, and password cracking, which are widely used to gain unauthorized access to admin panels. It also points out that weak authentication practices and inadequate security monitoring significantly increase the chances of data breaches and financial losses.

The discussion emphasizes the importance of adopting layered security approaches in e-commerce platforms to minimize cyber risks and protect critical information. It recommends integrating secure authentication methods along with continuous monitoring systems within admin login portals to identify suspicious activities and block unauthorized access. The assessment determines that implementing advanced security measures is essential for maintaining safe and reliable e-commerce operations. This overview establishes a solid base for designing secure admin login systems and reinforces the importance of OTP authentication and cyber deception techniques in modern web environments.

2.2 C. ZHANG, HONEYFACTORY : CONTAINER BASED CYBER DECEPTION HONEYNET ARCHITECTURE[4]

This paper focuses on using honeypots and decoy environments to trap attackers and monitor their behaviour in real time. The authors explain that traditional security systems mainly focus on blocking attackers, but cyber deception techniques aim to mislead attackers and collect useful information about their actions. This helps in improving intrusion detection and strengthening system security.

The proposed approach introduces a scalable honeynet architecture that creates multiple fake environments using container technology. These decoy environments look like real systems and attract attackers, allowing security teams to monitor their activities without risking the real system. The paper explains that honeypots can record attacker behaviour, track intrusion patterns, and generate alerts for suspicious activities. The Honey Factory framework provides an efficient way to deploy and manage multiple honeypots in a secure and controlled environment.

The analysis highlights the advantages of cyber deception techniques in modern cybersecurity, including improved threat detection, attacker monitoring, and system protection. The authors also discuss how honeypots can reduce the risk of data breaches and enhance security monitoring in web applications and enterprise systems.

Overall, the findings indicate that cyber deception and honeynet architectures are effective in detecting cyber threats and improving overall system security. This paper supports the implementation of decoy dashboards and honeypot mechanisms in secure admin login systems and provides a strong reference for cyber deception-based security approaches.

2.3 Z. HUANG, IMPACT OF EXPOSED PASSWORDS ON HONEYWORD EFFICACY [5]

This approach focuses on the use of honeypots and decoy environments to trap attackers and monitor their behaviour in real time. The authors explain that traditional security mechanisms mainly concentrate on blocking intruders, whereas cyber deception techniques are designed to mislead attackers and collect valuable insights about their actions. This method improves intrusion detection and strengthens overall system security.

The proposed solution presents a scalable honeynet architecture that creates multiple fake environments using container-based technology. These decoy systems closely resemble real platforms and attract attackers, allowing security teams to observe their activities without putting the actual system at risk. It explains how honeypots can record attacker behaviour, identify intrusion patterns, and generate alerts for suspicious actions. The Honey Factory framework provides an effective way to deploy and manage multiple honeypots within a secure and controlled environment.

The assessment highlights the advantages of cyber deception techniques in modern cybersecurity, including enhanced threat detection, continuous monitoring of attackers, and stronger system protection. It also explains how honeypots help reduce the risk of data breaches and improve monitoring capabilities in web applications and enterprise systems.

Overall, the result indicates that cyber deception strategies and honeynet architectures are highly effective in identifying cyber threats and improving system security. This approach supports the use of decoy dashboards and honeypot-based mechanisms in secure admin login systems and acts as a valuable reference for deception-based security implementations.

2.4 Z. MORIĆ, V. DAKIĆ, AND D. REGVART, ADVANCING CYBERSECURITY WITH HONEYPOTS AND DECEPTION STRATEGIES [10]

This analysis explores advanced cybersecurity techniques using honeypots and deception strategies to protect sensitive systems from cyber attacks. It highlights the increasing demand for intelligent security mechanisms capable of detecting and preventing unauthorized access in modern digital environments. The authors explain that conventional security methods primarily focus on blocking attackers, whereas deception-based techniques are designed to mislead intruders and collect useful information about their activities.

The discussion examines different types of honeypots and their role in enhancing intrusion detection and system protection. Honeypots are decoy systems created to attract attackers and observe their behaviour without impacting the actual system. It explains that deception strategies enable security teams to study attacker techniques, identify system vulnerabilities, and strengthen defense mechanisms.

The evaluation also emphasizes the importance of continuous monitoring and logging of attacker activities to improve threat analysis and detection. It concludes that honeypots and deception-based approaches are effective in minimizing cyber threats and enhancing overall system security.

The findings highlight that integrating deception techniques into web applications and enterprise systems can provide stronger protection against cyber attacks. This analysis supports the implementation of decoy dashboards and attacker monitoring mechanisms in secure admin login systems and reinforces the concept of cyber deception used in the project.

2.5 S. S. ABDULSATTAR, ADAPTIVE DECEPTION FRAMEWORK FOR AI-DRIVEN CYBER ATTACKS [11]

An adaptive deception framework is presented to counter AI-driven cyber attacks and advanced hacking techniques. The work highlights the rising use of artificial intelligence in cyber threats and the need for intelligent security systems to safeguard sensitive data and digital platforms. The author explains that conventional security solutions are not sufficient to handle modern threats that rely on automation and machine learning.

The proposed model introduces an adaptive deception framework that integrates honeypots, intrusion detection systems, and alert mechanisms to identify and respond to cyber attacks in real time. It emphasizes the importance of combining multiple security layers to build a robust defense system. The framework is capable of automatically detecting suspicious activities, misleading attackers, and generating alerts to notify administrators about potential threats, thereby enhancing system security and reducing the risk of data breaches.

The assessment highlights that adaptive deception strategies play a crucial role in modern cybersecurity environments, especially for protecting web applications and enterprise systems. It concludes that intelligent deception frameworks significantly enhance intrusion detection and provide stronger protection against sophisticated cyber attacks.

This approach supports the integration of alert systems, continuous monitoring, and decoy-based protection in secure admin login systems, thereby strengthening the overall security architecture of the project.

2.6 P.DAS, A.BANERJEE, AND R.CHATTERJEE, INTELLIGENT INTRUSION DETECTION USING DECOY INTERFACES IN PLATFORMS [12]

A comparative analysis examines honeyword-based and honeypot-based authentication systems used in modern web security. It focuses on evaluating different authentication approaches and their effectiveness in detecting cyber attacks and protecting sensitive data. The authors explain that conventional login systems are highly vulnerable to password-based attacks and therefore require advanced security mechanisms to enhance protection.

The evaluation contrasts honeyword-based authentication, which uses decoy passwords to identify unauthorized access attempts, with honeypot-based systems that create fake environments to observe suspicious activities. It explains that honeywords help detect intrusions during login attempts, while honeypots assist in monitoring attacker behaviour and gathering valuable security insights.

By integrating both techniques, a robust and dependable security framework can be developed to safeguard web applications and admin panels. The discussion also highlights various honeyword generation methods that ensure fake passwords appear realistic, making it difficult for attackers to distinguish them from genuine ones. As a result, honeyword-based systems add an extra layer of security and help prevent unauthorized access to web applications and administrative interfaces.

CHAPTER 3

EXISTING SYSTEM

3.1 EXISTING SYSTEM

The existing system, security is primarily based on traditional username and password authentication methods, which provide only basic protection against unauthorized access. Some systems incorporate honeywords (decoy passwords) to detect credential misuse; however, when a honeyword is used, the system typically only generates an alert without taking any active measures to prevent further intrusion. The real admin login page remains directly accessible, allowing attackers to repeatedly attempt login using stolen or guessed credentials.

Additionally, there is no mechanism to divert or mislead attackers away from the actual system. Once access is attempted, the system does not provide any form of interaction or trapping mechanism to study attacker behavior. There is also a lack of monitoring and detailed logging of malicious activities, which limits the ability to analyze threats and improve security measures.

Overall, the existing system offers protection only at the password verification level and does not secure the admin panel or database effectively. Even if a security breach is detected, the real system components remain exposed, making it vulnerable to data theft and unauthorized access.

Some advanced systems introduce honeywords (decoy passwords) to detect password breaches. However, these systems are mostly limited to detection only. When a honeyword is used, the system generates an alert but does not take any strong action such as blocking access or misleading the attacker. This means that even after detecting suspicious activity, the real admin login interface remains visible and accessible.

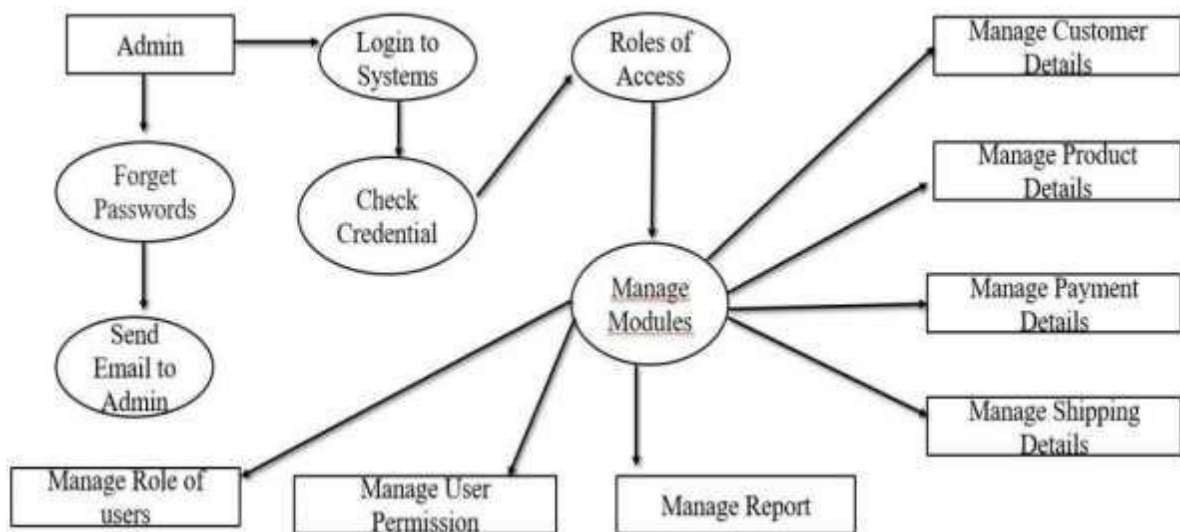


Figure 3.1 Existing System

3.2 DISADVANTAGES

- Relies only on username and password authentication, which is vulnerable to attacks.
- Cannot distinguish between a real user and attacker if correct credentials are used.
- Honeyword-based systems only generate alerts without taking preventive action.
- Real admin login page remains visible and accessible to attackers.
- No mechanism to divert or mislead attackers away from the real system.
- No use of decoy systems (honeypots) to trap malicious users.
- Lack of proper monitoring and tracking of attacker activities.
- No detailed logging of login attempts, IP address, or user behavior.

CHAPTER 4

PROBLEMS IDENTIFIED

The modern E-Commerce systems, the admin panel plays a critical role in managing business operations such as product updates, order processing, customer data handling, and financial transactions. However, most existing admin login systems are built using traditional authentication techniques that rely only on username and password verification. This approach creates several security vulnerabilities, making the system highly prone to cyber attacks and unauthorized access.

One of the major problems identified is the weak authentication mechanism used in existing systems. Since passwords are often predictable or reused across multiple platforms, attackers can easily guess or obtain them through phishing, keylogging, or data breaches. Once the password is compromised, there is no additional verification layer to stop the attacker from accessing the system.

Another critical issue is the system's susceptibility to brute-force and dictionary attacks. In many cases, attackers can attempt multiple login combinations without any strict limitation or monitoring. The absence of account lockout mechanisms, CAPTCHA verification, or login attempt restrictions allows attackers to continuously try different passwords until they successfully gain access. This significantly increases the risk of unauthorized entry into the admin panel.

The lack of multi-factor authentication (MFA) is also a major problem. Without OTP or secondary verification methods, the system cannot confirm the identity of the user. Even if the correct credentials are entered, there is no guarantee that the user is legitimate. This creates a security gap where attackers can easily exploit stolen credentials and access sensitive information.

Another important issue is the absence of intrusion detection and activity monitoring systems. Existing systems do not track login attempts, user behavior, or suspicious activities. There is no mechanism to analyze whether a login attempt is normal or malicious. This lack of monitoring makes it difficult to detect cyber attacks at an early stage and prevents administrators from taking timely action.

Furthermore, traditional systems do not implement cyber deception techniques such as honeywords and honeypots. Honeywords are fake passwords that help in detecting attackers, while honeypots are decoy environments designed to trap and monitor malicious users. Without these techniques, attackers who gain access can directly interact with the real system, causing serious damage. There is no way to mislead attackers or collect information about their behavior for further security improvements.

Another major problem is the absence of real-time alert and notification systems. Administrators are not informed about suspicious login attempts, multiple failed logins, or unauthorized access. This delay in detection increases the impact of cyber attacks, as attackers can operate within the system without being noticed. Immediate alerts are essential for quick response and prevention of further damage.

In addition, insecure password recovery mechanisms contribute to system vulnerability. Many existing systems rely on weak verification methods such as simple email reset links or security questions, which can be easily exploited. Attackers can use these methods to reset admin passwords and gain unauthorized access to the system.

CHAPTER 5

PROPOSED SYSTEM

The given diagram shows the working structure of a traditional E- Commerce admin system, where the administrator performs various operations through a centralized platform. This system mainly focuses on managing business activities such as user data, product information, payments, and shipping details. However, it follows a basic workflow without advanced security mechanisms.

The process begins with the Admin who acts as the primary user of the system. The admin accesses the platform through the Login to Systems module by entering login credentials such as username and password. These credentials are then validated in the Check Credential stage, where the system verifies the entered data with the stored database records. If the credentials match, the admin is granted access to the system; otherwise, access is denied.

After successful authentication, the system assigns Roles of Access to the admin. This module determines what actions the admin can perform based on predefined permissions. For example, different admins may have different roles such as super admin, manager, or staff, each with specific access rights. This role- based access control helps in organizing system operations but does not provide strong security against unauthorized access.

The system includes a Forget Password feature, which is used when the admin is unable to remember login credentials. In this process, a password reset request is initiated, and the system sends a recovery link or password information through the Send Email to Admin module. This allows the admin to reset the password and regain access to the system.

Despite providing essential functionalities, the existing system has several limitations. The login process relies only on basic credential verification without additional security layers such as OTP or multi-factor authentication. There is no mechanism to detect unauthorized login attempts or prevent brute-force attacks. Furthermore, the system does not include cyber deception techniques like honeywords or honeypots, which could help in identifying attackers. The absence of real-time alerts and monitoring makes it difficult to track suspicious activities.

The existing system is functional but lacks advanced security features, making it vulnerable to modern cyber threats. This highlights the need for an improved system that incorporates strong authentication, intrusion detection, and decoy protection mechanisms.

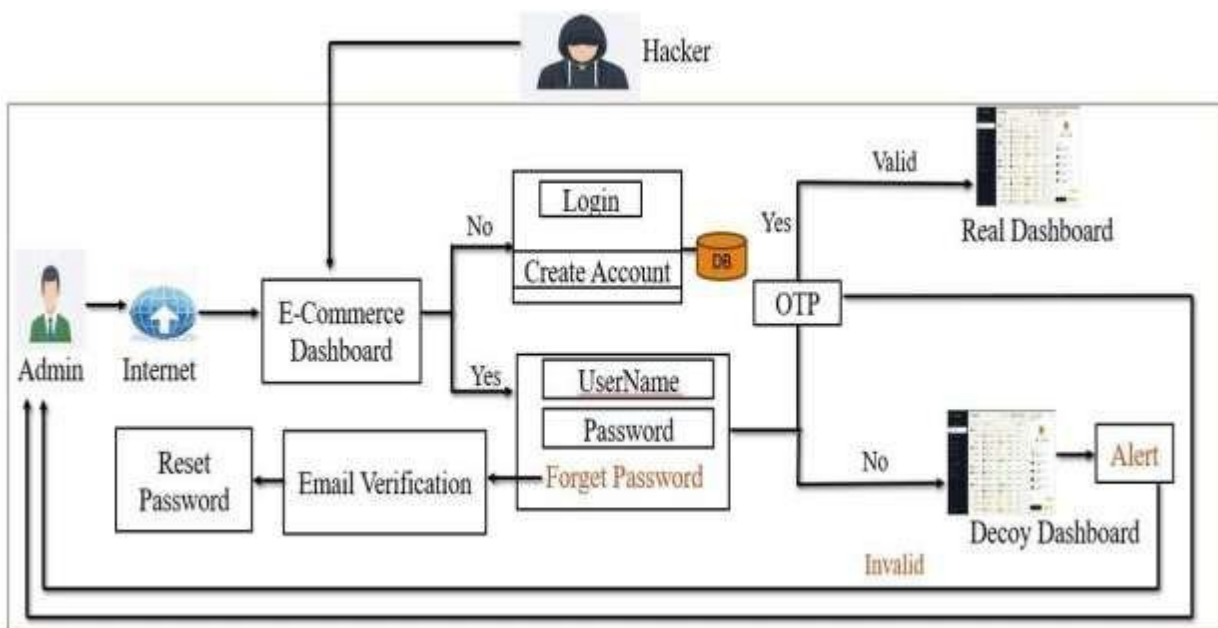


Figure.5.1 Proposed System

CHAPTER 6

SYSTEM REQUIREMENTS

6.1 HARDWARE REQUIREMENTS

Processor	:	Intel Core i3 or Equivalent
RAM	:	Minimum 4 GB (8 GB Recommended)
Storage	:	Minimum 250 GB Hard
Disk System Type	:	32-bit or 64-bit
Input Devices	:	Keyboard and Mouse
Internet	:	Stable Internet Connection Required

6.2 SOFTWARE REQUIREMENTS

Operating System	:	Windows 10 / Windows 11
Frontend	:	HTML, CSS, JavaScript
Development Tool	:	Visual Studio Code (VS Code)
Web Browser	:	Google Chrome / Microsoft Edge
Third-Party Tool	:	EmailJS (for OTP & Alert Notifications)
Storage	:	Local Storage / Simple Database

CHAPTER 7

SYSTEM IMPLEMENTATIONS

7.1 LIST OF MODULES

The proposed system is divided into several functional modules to ensure efficient operation, security, and maintainability. Each module is designed to handle a specific task within the admin login system, such as user authentication, password management, and intrusion detection. This modular approach improves system organization, simplifies development, and enhances overall performance. It also enables better security by separating critical functionalities like real and decoy access control.

- Login Portal Module
- Authentication Module
- Password Recovery Module
- Real Admin Dashboard Module
- Decoy Dashboard Module
- Alert Module

7.2 MODULES DESCRIPTION

The system is designed using a modular architecture to ensure a high level of security, flexibility, and ease of maintenance. In this approach, the entire application is divided into multiple independent modules, each responsible for handling a specific function within the admin login system. This separation of concerns allows developers to manage, update, and debug each component without affecting the overall system, thereby improving reliability and scalability. Each module plays a critical role in the system's operation. For instance, the login handling module is responsible for capturing user credentials and securely transmitting them for verification.

The authentication module validates the credentials against stored data using secure hashing and encryption techniques. The password recovery module provides a secure mechanism for administrators to regain access to their accounts, typically involving identity verification and controlled password reset procedures. Additionally, the attack detection module continuously monitors login attempts and user behavior to identify suspicious activities such as brute-force attacks, unusual login patterns, or repeated failed attempts.

7.2.1 Login Portal Module

The Login Portal Module acts as the front-end interface through which the admin interacts with the system. It is responsible for capturing user credentials such as username and password in a structured format. The module is designed with proper validation techniques to ensure that all required fields are filled correctly before submission. It prevents invalid inputs, such as empty fields or incorrect formats, thereby reducing unnecessary processing in the backend.

Additionally, the module provides features like “Forgot Password” and email verification links, improving usability. It also ensures secure data transmission by preventing exposure of sensitive information during input. The design of this module focuses on simplicity and accessibility so that users can easily navigate and perform login operations.

7.2.2 Authentication Module

The Authentication Module is the most critical part of the system as it handles identity verification and access control. It compares the entered credentials with the stored data in the database and determines whether the user is authorized. This module implements multi-factor authentication by generating and validating an OTP, ensuring that even if the password is compromised, unauthorized access is prevented.

7.2.3 Password Recovery Module

The Password Recovery Module ensures that users can regain access to their account in case they forget their password. It follows a secure process that verifies the identity of the user before allowing password reset. When the admin selects the password recovery option, they must provide their registered email address.

The system then sends an OTP or a password reset link to the email. The module verifies the OTP within a limited time period to ensure security. Once verified, the user is allowed to create a new password. The system enforces strong password policies, including minimum length and complexity requirements, to prevent weak passwords.

Additionally, all password recovery attempts are logged for monitoring suspicious activities. This module balances both security and convenience, ensuring that legitimate users can recover their accounts without compromising system safety.

7.2.4 Real Admin Dashboard Module

The Real Admin Dashboard Module is the central control panel of the system, accessible only after successful authentication. It provides all functionalities required for managing the e-commerce platform. The admin can perform operations such as adding, updating, and deleting products, managing customer information, handling payments and shipping details, and generating reports.

This module interacts directly with the database to perform CRUD operations efficiently. It is designed with role-based access control, ensuring that only authorized users can perform specific actions. The interface is user-friendly and organized, allowing easy navigation between different sections.

Security measures such as session management and access restrictions are implemented to prevent unauthorized usage. This module ensures smooth and efficient management of the entire e-commerce system.

7.2.5 Decoy Dashboard Module

The Decoy Dashboard Module is an innovative security feature designed to handle unauthorized access attempts. When a honeyword is entered, the system redirects the user to this fake dashboard instead of denying access. The dashboard is designed to look identical to the real admin panel, making the attacker believe that they have successfully logged in.

However, this environment is completely isolated from the real system and does not provide access to actual data. All activities performed by the attacker are recorded, including clicks, navigation patterns, and attempted actions. This information can be used to analyze attack behavior and improve security strategies.

The module plays a significant role in delaying attackers and protecting the real system from being compromised. It also helps in gathering intelligence about potential threats.

7.2.6 Alert Module

The Alert Module is responsible for monitoring system activities and notifying the admin about suspicious behavior. When events such as honeyword login attempts, repeated failed logins, or unusual access patterns are detected, the module generates alerts. These alerts are sent via email using integrated services like EmailJS.

The module ensures real-time communication with the admin, allowing immediate action to prevent further attacks. It also maintains logs of all alerts and suspicious activities for future reference. This enhances the overall security of the system by ensuring that no malicious activity goes unnoticed.

CHAPTER 8

SYSTEM TESTING

8.1 UNIT TESTING

Unit testing is the process of testing individual components or modules of the system independently to ensure that each unit performs its intended function correctly. In this project, every module such as the login form, OTP generation, input validation, and password reset functionality is tested separately before integration. This helps in identifying errors at the earliest stage of development.

For example, the login module is tested to verify that it correctly accepts valid usernames and passwords while rejecting invalid or empty inputs. Similarly, the OTP generation function is tested to ensure that it produces unique and time-bound codes, and that the verification logic works accurately. Unit testing is usually performed by developers using test cases for each function.

This type of testing improves code quality, reduces debugging time, and ensures that each module is reliable before combining it with others. It acts as the foundation for all other testing processes, as a system with properly tested units is less likely to fail during integration.

8.2 INTEGRATION TESTING

Integration testing focuses on testing the interaction between different modules to ensure that they work together seamlessly. After individual modules are tested, they are combined, and their communication is verified. In this project, integration testing ensures that the login portal correctly sends user data to the authentication module and that the authentication module processes the data and redirects users appropriately.

For instance, when a user logs in, the credentials must flow from the login module to the authentication module, which then validates them and determines whether to grant access or redirect to the decoy dashboard. Similarly, the password recovery module must successfully interact with the email service to send OTPs, and the alert module must communicate with the notification system to send alerts.

8.3 SYSTEM TESTING

System testing is the process of evaluating the complete and integrated system to verify that it meets all specified requirements. It is performed after all modules have been integrated and focuses on testing the system as a whole rather than individual parts. In this project, system testing ensures that the entire workflow—from login to dashboard access operates correctly and securely.

For example, it verifies the complete login process, including entering credentials, OTP verification, and access to the appropriate dashboard. It also checks the proper functioning of the decoy dashboard when a honeyword is entered and ensures that alerts are triggered for suspicious activities.

System testing validates both functional requirements (such as correct output) and non-functional requirements (such as performance and security). It ensures that the system is ready for deployment and can handle real-world scenarios effectively.

8.4 SECURITY TESTING

Security testing is one of the most important aspects of this project, as it focuses on protecting the system from unauthorized access and cyber threats. It involves identifying vulnerabilities and ensuring that all security mechanisms are functioning correctly.

In this project, security testing includes verifying OTP-based authentication, ensuring that only authorized users can access the system. It also tests the honeyword mechanism by entering fake passwords and checking whether the system correctly redirects users to the decoy dashboard. The system is tested against brute-force attacks by attempting multiple incorrect logins and verifying that access is restricted after a certain number of attempts.

Additionally, session management is tested to ensure that user sessions are secure and cannot be hijacked. The alert module is also tested to confirm that it sends notifications to the admin whenever suspicious activity is detected.

8.5 USER ACCEPTANCE TESTING

User Acceptance Testing is the final level of testing performed by end users or clients to ensure that the system meets their requirements and expectations. It focuses on usability, functionality, and overall user experience. In this project, the admin tests the system to verify that all features are easy to use and function correctly.

For example, the admin checks whether the login process is simple and intuitive, whether the dashboard is easy to navigate, and whether all operations such as managing products and users are performed smoothly. The password recovery process is also tested to ensure that it is convenient and secure.

Feedback from users is collected during this stage, and any necessary improvements are made before deployment. This testing ensures that the system is practical, user-friendly, and suitable for real-world usage.

8.6 PERFORMANCE TESTING

Performance testing evaluates how efficiently the system operates under normal and expected conditions. It focuses on response time, speed, scalability, and system stability. In this project, performance testing ensures that the login process is fast and does not cause delays for users. It also checks how quickly OTPs are generated and delivered to the user's email.

The loading time of both the real admin dashboard and the decoy dashboard is measured to ensure smooth navigation. The system is also tested for its ability to handle multiple requests without slowing down. This testing helps identify performance bottlenecks and ensures that the system provides a fast, reliable, and smooth user experience.

CHAPTER 9

RESULT AND DISCUSSION

The result and discussion section presents the E-Commerce Admin Login with Decoy Protection system was successfully implemented and tested. The results show that the system effectively enhances security using OTP authentication, honeyword detection, and a decoy dashboard.

The login and authentication modules worked correctly by validating credentials and granting access to the real admin dashboard for valid users. When a honeyword was entered, the system successfully redirected the user to the decoy dashboard, helping to detect unauthorized access without exposing real data.

The alert module generated notifications for suspicious login attempts, ensuring real-time monitoring. The password recovery module also functioned properly by allowing secure password reset through email verification.

Overall, the system performed efficiently with quick response time and smooth operation. The results confirm that the proposed system provides better security compared to traditional login systems and successfully protects the admin panel from potential threats.

CHAPTER 10

CONCLUSION AND FUTURE WORK

10.1 CONCLUSION

In this project, a deep learning-based system for the detection and classification of gastrointestinal lesions from endoscopic images has been successfully developed. The main objective of the system is to assist healthcare professionals in identifying abnormalities within the gastrointestinal tract more efficiently and accurately. Gastrointestinal diseases such as ulcers, polyps, and bleeding regions can lead to serious health complications if not detected at an early stage. Therefore, the development of automated diagnostic systems plays an important role in improving medical diagnosis and patient care.

The proposed system utilizes deep learning techniques, particularly Convolutional Neural Networks (CNNs), to analyse endoscopic images and extract meaningful features associated with gastrointestinal abnormalities. The system follows a structured workflow consisting of several stages, including image acquisition, image preprocessing, feature extraction, lesion classification, and result visualization. Each module contributes to the overall functionality of the system and ensures accurate detection of gastrointestinal lesions.

Experimental results show that the proposed deep learning model is capable of identifying abnormal regions in endoscopic images with a high level of accuracy. The system successfully processes medical images and classifies them into different categories based on the detected lesion type. The use of deep learning techniques enables the system to analyse complex image patterns that may not be easily detected through manual examination.

10.2 FUTURE WORK

Although the proposed system demonstrates promising results, several improvements can be made in the future to enhance its performance and applicability in real-world healthcare environments. One possible enhancement is the expansion of the training dataset. By incorporating larger and more diverse datasets from multiple medical institutions, the deep learning model can learn a wider variety of gastrointestinal conditions and improve its generalization capability.

Another potential improvement is the integration of real-time lesion detection during endoscopic procedures. Real-time analysis would allow clinicians to receive immediate feedback while examining the gastrointestinal tract, enabling faster identification of abnormalities. This capability could significantly improve the efficiency of endoscopic examinations.

Future research may also explore the use of advanced deep learning architectures such as ResNet, EfficientNet, or transformer-based models to further improve classification accuracy. These models have shown strong performance in medical image analysis and may enhance the ability of the system to detect subtle abnormalities.

In addition, explainable artificial intelligence techniques can be incorporated into the system to provide better interpretation of the model's predictions. Visualization methods such as heatmaps and attention maps can highlight the regions of the image that influence the model's decision, helping clinicians understand and trust the system's output. Another future enhancement involves integrating the system with hospital information systems and telemedicine platforms.

APPENDIX - A

SOURCE CODE

HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>ShopNova - Premium E-Commerce Store</title>
<link
  href="https://fonts.googleapis.com/css2?family=Clash+Display:wght@400;500;600;
  700&family=Plus+Jakarta+Sans:wght@300;400;500;600;700;800&display=swap"
  rel="stylesheet">
<style>
:root {
  --primary: #f97316;
  --primary-dark: #ea580c;
  --secondary: #ec4899;
  --accent: #8b5cf6;
  --success: #10b981;
  --warning: #f59e0b;
  --danger: #ef4444;
}
* { margin: 0; padding: 0; box-sizing: border-box; }
html { scroll-behavior: smooth; }
body { font-family: 'Plus Jakarta Sans', sans-serif; background: var(--dark); color:
  var(--text); overflow-x: hidden; }
```

```

.navbar {
  position: fixed; top: 0; left: 0; right: 0; z-index: 1000;
  display: flex; align-items: center; justify-content: space-between;
  padding: 16px 5%;
  background: rgba(10,10,26,0.92);
  backdrop-filter: blur(20px);
  border-bottom: 1px solid var(--border);
}
.logo { font-family: 'Clash Display', sans-serif; font-size: 26px; font-weight: 700;
  background: linear-gradient(135deg, var(--primary), var(--secondary)); -webkit-
  background-clip: text; -webkit-text-fill-color: transparent; text-decoration: none;
  cursor: pointer; }
.cart-badge { position: absolute; top: -6px; right: -6px; background: var(--primary);
  color: white; width: 18px; height: 18px; border-radius: 50%; font-size: 10px; display:
  flex; align-items: center; justify-content: center; font-weight: 700; transition:
  transform 0.2s; }
.cart-badge.bump { transform: scale(1.5);
}.....
</body>
</html>

```

Database.js

```

const sqlite3 = require('sqlite3').verbose();
const bcrypt = require('bcryptjs');
const path = require('path');
const DB_PATH = path.join(__dirname, 'admin.db');
const db = new sqlite3.Database(DB_PATH, (err) => {
  if (err) console.error('DB Connection Error:', err);
  else console.log('■ SQLite Database Connected');
});
function initDatabase() {
  db.serialize() => {
    // Admin Users Table

```

```

is_active INTEGER DEFAULT 1,
failed_attempts INTEGER DEFAULT 0,
last_failed_attempt DATETIME,
created_at DATETIME DEFAULT CURRENT_TIMESTAMP
)`);

// OTP Table
db.run(`CREATE TABLE IF NOT EXISTS otps (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  email TEXT NOT NULL,
  otp TEXT NOT NULL,
  type TEXT NOT NULL,
  expires_at DATETIME NOT NULL,
  used INTEGER DEFAULT 0,
  created_at DATETIME DEFAULT CURRENT_TIMESTAMP
)`);

// Security Logs (for decoy tracking)
db.run(`CREATE TABLE IF NOT EXISTS security_logs (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  ip_address TEXT,
  username_attempted TEXT,
  user_agent TEXT,
  device_info TEXT,
  login_time DATETIME DEFAULT CURRENT_TIMESTAMP,
  action_type TEXT DEFAULT 'decoy_access',
  pages_visited TEXT,
  session_duration INTEGER DEFAULT 0,
  threat_level TEXT DEFAULT 'HIGH'
)`);

// Sessions Table
db.run(`CREATE TABLE IF NOT EXISTS sessions (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  admin_id INTEGER,
  token TEXT UNIQUE NOT NULL,
  ip_address TEXT,

```

```

    created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
    expires_at DATETIME NOT NULL,
    FOREIGN KEY(admin_id) REFERENCES admins(id));
// Create default admin if not exists
db.get("SELECT id FROM admins WHERE username = 'admin'", async (err, row) =>
{
  if(!row) {
    const hashedPassword = await bcrypt.hash('Admin@123', 12);
    db.run(
      `INSERT INTO admins (username, email, password, full_name, role) VALUES
      (?, ?, ?, ?, ?)`,
      ['admin', 'rohinidharmaraj34@gmail.com', hashedPassword, 'Super Admin',
      'superadmin'],
      (err) => {
        if(!err) console.log(' Default admin created: username=admin,
password=Admin@123');
      }
    );
  }
});
console.log(' Database tables initialized');
});
}
module.exports = { db, initDatabase };

```

emailService.js

```

const nodemailer = require('nodemailer');
const createTransporter = () => {
  return nodemailer.createTransport({
    service: 'gmail',
    auth: {
      user: process.env.EMAIL_USER,
      pass: process.env.EMAIL_PASS
    }
  })
}

```

```

const generateOTP = () => {
  return Math.floor(100000 + Math.random() * 900000).toString();
};

const sendOTPEmail = async (toEmail, otp, type = 'login') => {
  const subject = type === 'login'
    ? '🛒 ShopNova Admin - Login Verification OTP'
    : '🔑 ShopNova Admin - Password Reset OTP';

  const htmlContent = `
    <!DOCTYPE html>
    <html>
    <head>
      <meta charset="utf-8">
      <style>
        body { font-family: Arial, sans-serif; background: #0a0a1a; margin: 0; padding:
        20px; }
        .container { max-width: 500px; margin: 0 auto; background: #1a1a2e; border-
        radius: 20px; overflow: hidden; }
        .header { background: linear-gradient(135deg, #f97316, #ec4899); padding: 30px;
        text-align: center; }
        .header h1 { color: white; margin: 0; font-size: 24px; font-weight: 800; }
        .body { padding: 40px 30px; }
        .otp-box { background: linear-gradient(135deg, #f97316, #ec4899); border-radius:
        16px; padding: 30px; text-align: center; margin: 25px 0; }
        .otp-code { font-size: 48px; font-weight: 900; letter-spacing: 10px; color: white;
        font-family: monospace; }
      </style>
    </head>
    <body>
      <div class="container">
        <div class="header">
          <h1>🛒 ShopNova Admin</h1>

```

```

</div>
<div class="body">
  <div class="info">
    <strong style="color:#e2e8f0;font-size:18px;">
      ${type === 'login' ? '🔑 Login Verification' : '🔒 Password Reset'}
    </strong>
    <br><br>

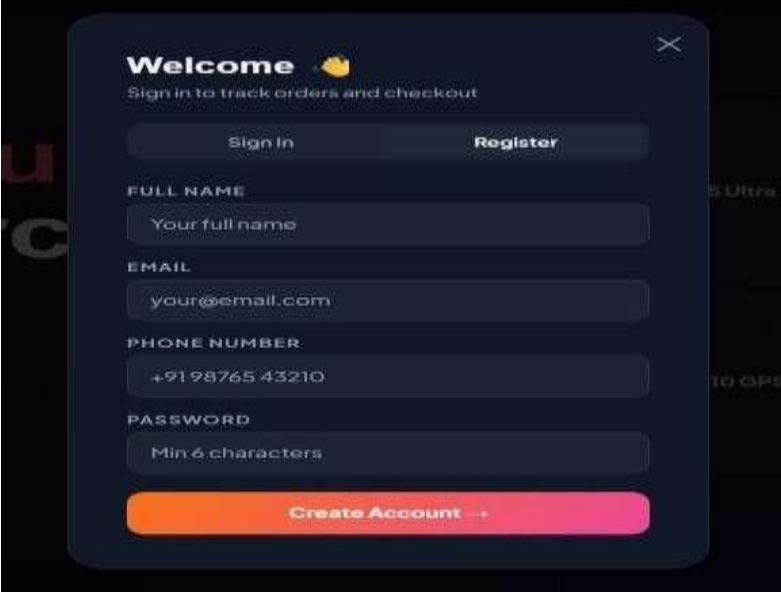
    Your One-Time Password is:
  </div>
  <div class="otp-box">
    <div class="otp-code">${otp}</div>
    <div class="otp-label">⌚ Expires in 10 minutes</div>
  </div>
  <div class="warning">
    .! Never share this OTP with anyone. If you did not request this, please secure your
    account immediately.
  </div>
</div>
<div class="footer">
  © 2026 ShopNova Admin Panel. Automated security email.
</div>
</div>
</body>
</html>
`;
try {
  const transporter = createTransporter();
  const info = await transporter.sendMail({
    from: `"ShopNova Security" <${process.env.EMAIL_USER}>`,
    to: toEmail,
    subject,
    html: htmlContent
  });
}

```

APPENDIX – B

SCREENSHOTS

1. Login Portal Module



A screenshot of a user login form titled "Welcome" with a yellow hand icon. Below the title is the instruction "Sign in to track orders and checkout". There are two buttons: "Sign In" and "Register". The form includes input fields for "FULL NAME" (placeholder: "Your full name"), "EMAIL" (placeholder: "your@email.com"), "PHONE NUMBER" (placeholder: "+91 98765 43210"), and "PASSWORD" (placeholder: "Min 6 characters"). At the bottom is a large orange-to-pink gradient button labeled "Create Account →".

Figure B.1.1 User Login



A screenshot of an admin login form titled "Admin Login" with a yellow padlock icon. Below the title is the instruction "Sign in to access the admin dashboard". The form includes input fields for "Username" (placeholder: "Enter username") and "Password" (placeholder: "Enter password"). There is a link "Forgot password?" in orange text. At the bottom is a large orange-to-pink gradient button labeled "Login →". Below the button is the text "Don't have an account? Request Access" in orange.

Figure B.1.2 Admin Login

2. Authentication Module

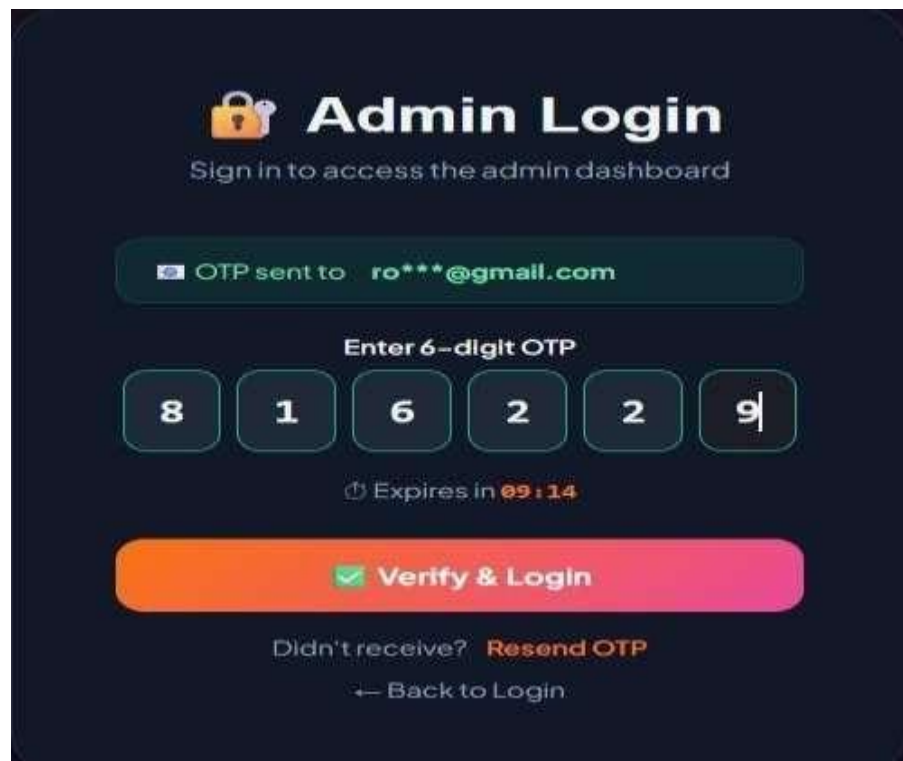


Figure B.2.1 Entering OTP

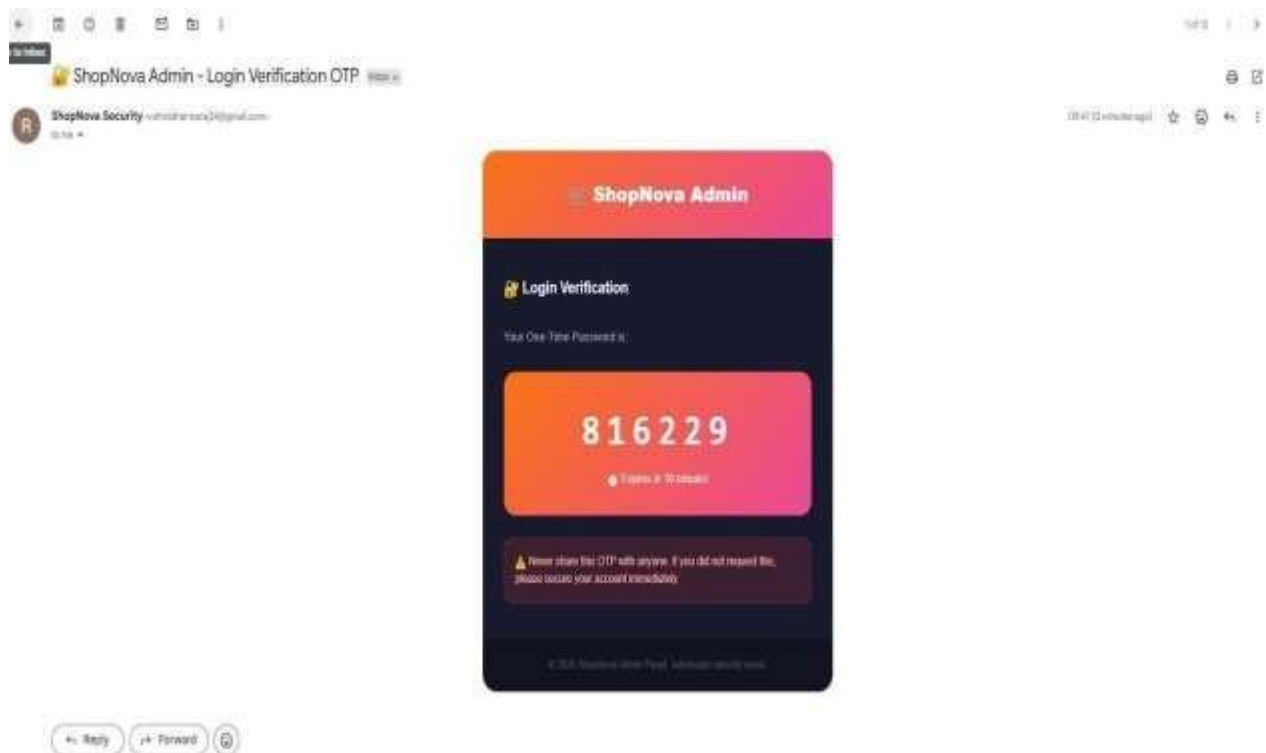


Figure B.2.2 OTP send to email

3. Password Recovery Module



The image shows a dark-themed 'Admin Login' screen. At the top, there's a lock icon and the text 'Admin Login' with a subtitle 'Sign in to access the admin dashboard'. Below this, a 'Reset Password' section is highlighted with a key icon. It instructs the user to 'Enter your registered email to receive a reset OTP'. A text input field labeled 'Admin Email' contains the email 'rohinidharmaraj34@gmail.com'. Below the input field is a large orange button labeled 'Send Reset OTP' and a smaller link that says '← Back to Login'.

Figure B.3.1 Enter email



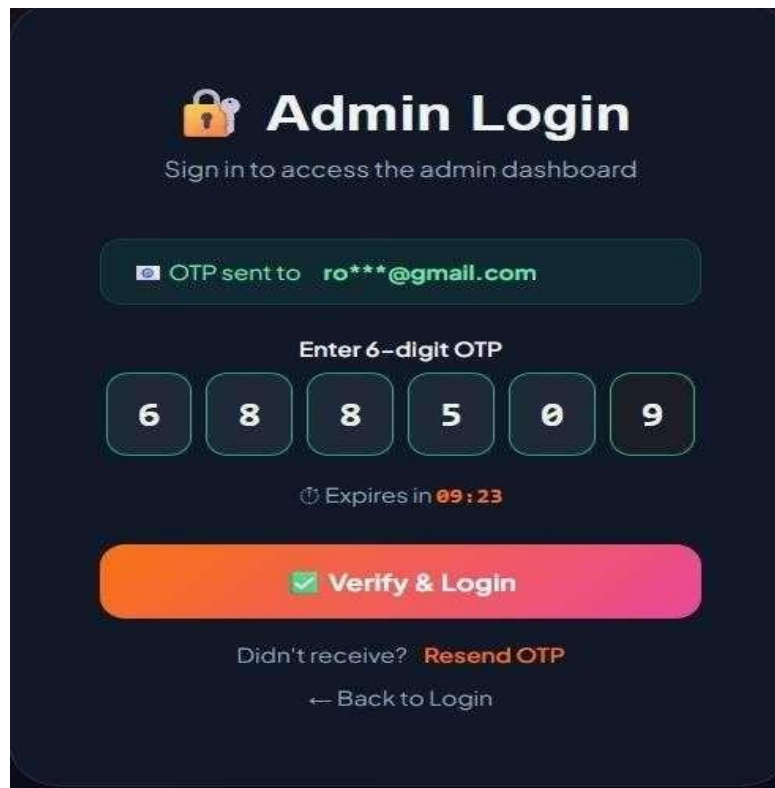
Figure B.3.2 OTP send to email



The image shows the same 'Admin Login' screen, but now the 'New Password' section is active. It instructs the user to 'Enter the OTP from your email and set a new password'. There are three input fields: 'OTP Code' (containing '430880'), 'New Password' (with masked dots), and 'Confirm Password' (also with masked dots and a toggle icon). A large orange button labeled 'Reset Password' is at the bottom.

Figure B.3.3 Reset Password

4. Real Admin Dashboard Module



The Admin Login screen features a dark blue background. At the top, there is a lock icon and the text "Admin Login" in white, followed by "Sign in to access the admin dashboard". Below this, a green box displays "OTP sent to ro***@gmail.com". A prompt "Enter 6-digit OTP" is followed by six input fields containing the digits 6, 8, 8, 5, 0, and 9. A timer indicates "Expires in 09:23". A large orange button with a checkmark icon says "Verify & Login". At the bottom, there are links for "Didn't receive? Resend OTP" and "← Back to Login".

Figure B.4.1 Enter OTP if enter password correct

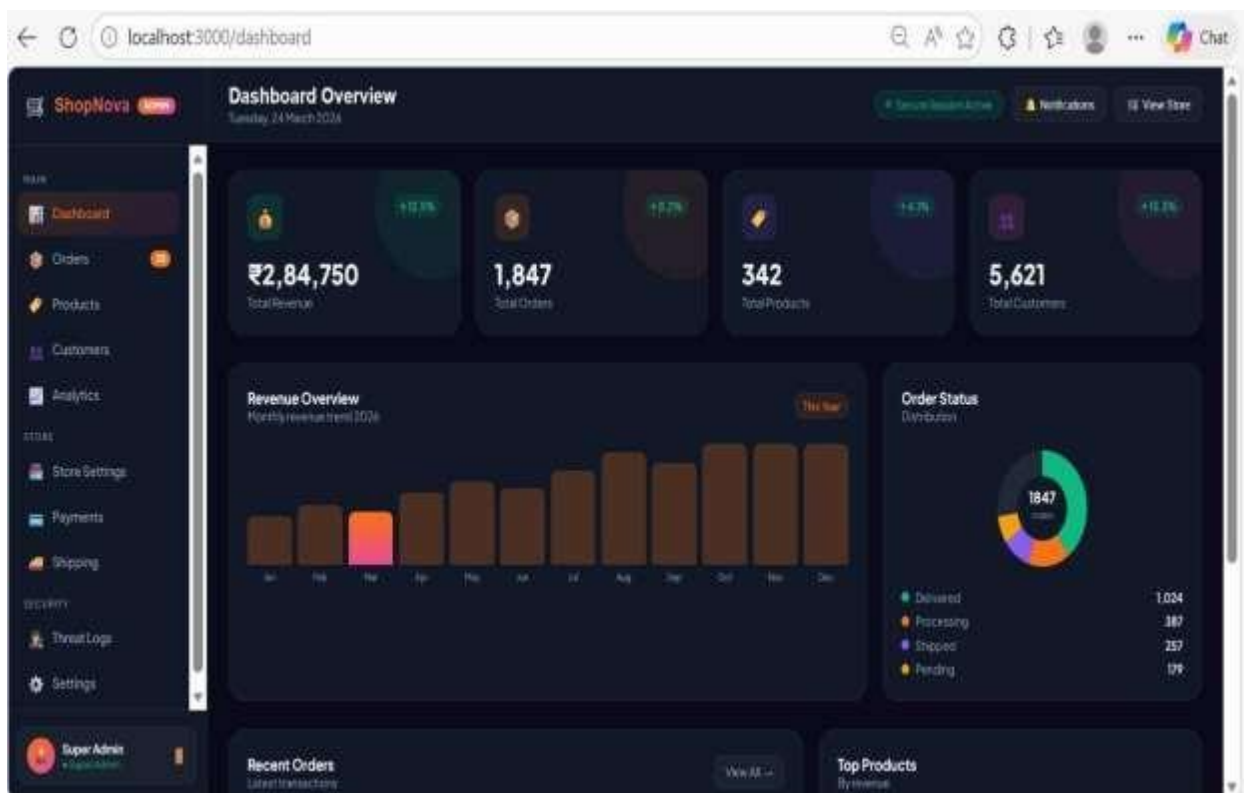


Figure B.4.2 Real Dashboard

5. Decoy Dashboard Module



Figure B.5.1 Invalid attempts

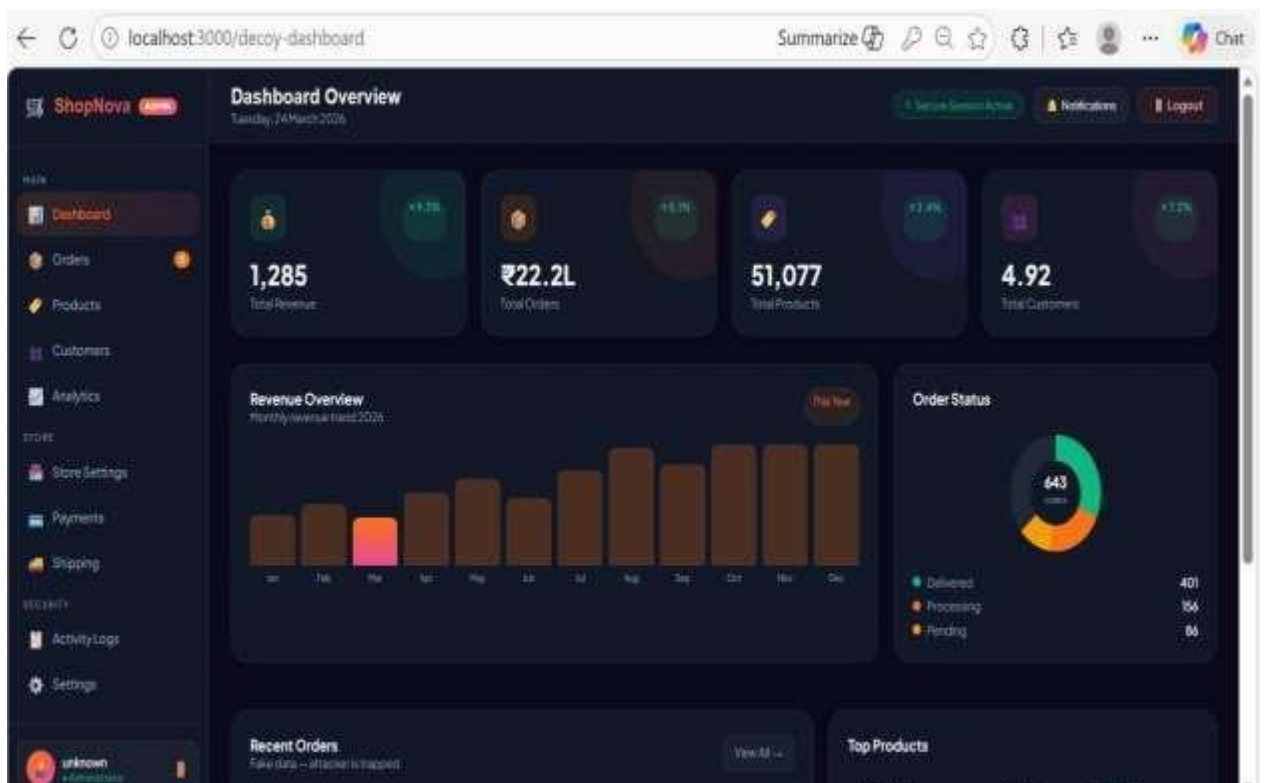


Figure B.5.2 Decoy Dashboard

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